Coursework and Project Cover Sheet

Third Year Undergraduate

Department of Civil and Environmental Engineering

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Module_____CIVE70004 Coastal Engineering_____

Assignment_____Wave Analysis Coursework_____

Supervisor_____ Dr Ioannis Karmpadakis _____

Deadline_____23/02/2026_____

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1. Introduction

Accurate prediction of wave-induced kinematics is fundamental to the reliable design of coastal and offshore structures. This report evaluates the horizontal velocity profile, u(z), beneath the largest wave crest recorded by a buoy in a large winter storm in a water depth of d=30 m. To determine the most appropriate framework for design criteria, this study compares five distinct wave models: Linear Regular, Stokes' Second & Fifth Order Regular, Linear Random, and Second-Order Random Wave theories.

2. Methodology and Theoretical Framework

To evaluate the velocity profile beneath the crest, different wave analysis approach will be used. In the process of analysis, the potential could be evaluated, and the velocity is the distance derivative of potential ($u = \frac{\partial \phi}{\partial x}$). For most analysis, velocity is the function of time and distance $u(x,t)$, thus $x = 0$ (the wave buoy is static), and the largest crest occurs within $-5 < t < 5$, the wave matching approach will be used to quantify the error of different analysis method by comparing η_{max} with the recorded data.

2.1 Regular Wave Theories

Regular wave theories approximate a sea state using a single, constant amplitude and frequency. To apply these theories to an extreme storm event, a representative Wave Height (H) and Period (T) must be extracted, and wave number(k) will be obtained iteratively by dispersion equation. Because maximum structural loading occurs at the absolute crest, this study defines T via three different method (zero up-crossing, zero down-crossing, crest-to crest and trough-to-trough) and to the difference between maximum and minimum observed crest elevations ($\eta_{max} - \eta_{min}$).

- Linear Regular Wave Theory (Free Wave): Assumes infinitesimally small wave amplitudes ($a \ll d, a \ll \lambda$) and neglects all nonlinear boundary terms
- Stokes' Second-Order Theory (Bound Wave): Incorporates initial nonlinearities, resulting in wave asymmetry (steeper crests and flatter troughs) by adding a second-order bound harmonic with a phase velocity locked to the free wave ($c = \omega/k$). The kinematic and dynamic free surface boundary conditions are also applied.
- Stokes' Fifth-Order Theories: Further enhance accuracy for steep, near-breaking waves by employing a perturbation expansion series to satisfy the nonlinear free-surface boundary conditions.

2.2 Linear Random Wave Theory (LRWT)

A real sea state is irregular and is physically best represented as a superposition of multiple free wave components. The horizontal velocity $u(x,z,t)$ is defined as:

$$u(x,z,t) = \sum_{m=1}^{M} a_m \omega_m \frac{\left(cosh(k_m(z+d)\right)}{sinh(k_m d)} cos(k_m x - \omega_m t + \alpha_m)$$

While robust for mean water levels, LRWT suffers from "high-frequency contamination" near the free surface of a large wave. High-frequency waves are erroneously extrapolated above the Still Water Level (SWL), causing the $cosh(k(z+d))$ term to grow exponentially. To mitigate this, the empirical Wheeler Stretching method is applied in industry, adjusting the vertical coordinate z to a stretched coordinate z^*.

2.3 Second-Order Random Wave Theory

To accurately model extreme irregular waves, a second-order random wave model was applied. Unlike linear theory, this model mathematically resolves nonlinear wave-wave interactions, calculating the secondary "bound waves" that physically cause sharper crests and amplitude modulation.

However, raw buoy spectra do not distinguish between linear free waves and existing nonlinear bound waves. Using the full spectrum causes the model to treat bound waves as free waves, erroneously "double counting" the nonlinear interactions. To isolate pure free-wave energy, a bandpass filter was applied. Frequencies outside the $0.05\,Hz < f < 0.3\,Hz$ range (chosen base on the amplitude-frequency spectrum by filter out low-amplitude wave)

were neglected to eliminate existing sum-interacted and difference-interacted bound wave contamination before computing the second-order kinematics.

3. Result

3.1 Wave Profile Reconstruction & Crest Error

The maximum wave crest was identified at $t = 0s$ with an observed elevation of $\eta_{obs} = 6.37\ m$. The surface profiles are reconstructed with 5 different analyses. In regular wave analysis, the T and H are required to guess by different method and the best fitted period is picked by trough-trough method(comparison figures in appendix). Figure 3.1 and 3.2 illustrates the reconstructed surface profiles.

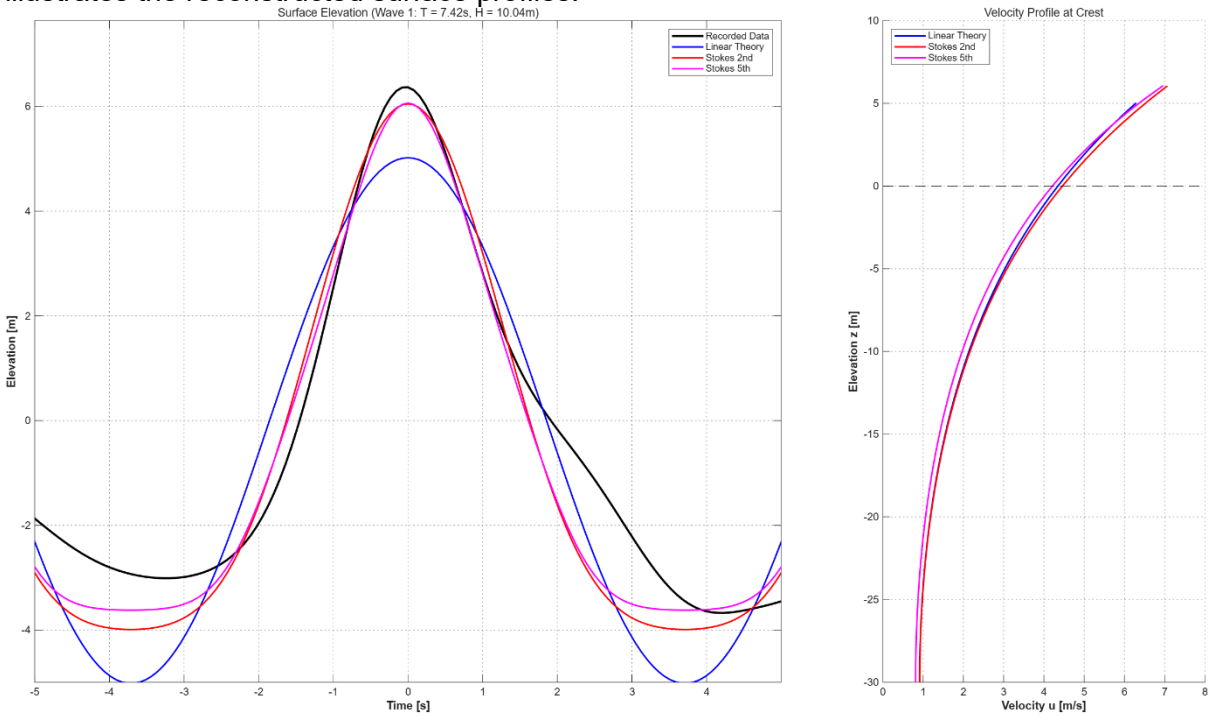


Figure 3.1 Surface profiles and velocity profiles for regular wave theories

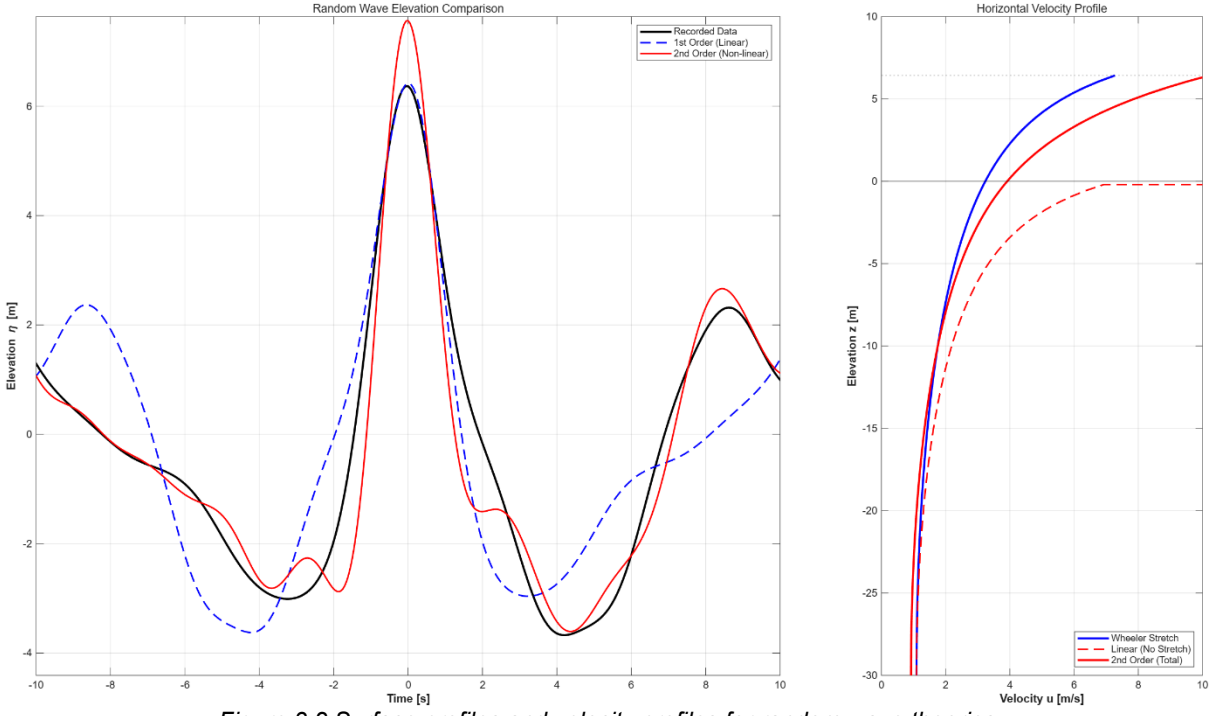


Figure 3.2 Surface profiles and velocity profiles for random wave theories

To evaluate the safety of each model for deck-clearance design, the crest height error $(\eta_{obs} - \eta_{model})$ was calculated in the table below:

Wave Mode	Modelled η_{max} (m)	Crest Error (m)	Error %
Linear Regular	5.02	1.346	21.14
Stoke's 2nd	6.05	0.316	4.96
Stoke's 5th	6.066	0.3	4.71
Linear Random	6.421	0.055	0.86
2nd Random	7.563	1.197	18.79

3.2 Horizontal Velocity Profiles

The depth variation in horizontal velocity, $u(z)$, was calculated directly beneath the highest crest $(x = 0)$ from the seabed $(z = -30m)$ to the respective free surface $(z = \eta_{max})$. Figures 3.1 and 3.2 above illustrate the velocity profile too.

The maximum horizontal velocities extracted at the free surface are:

- Linear Regular: $\mathrm{u_{max}} = 6.2884\ \mathrm{m/s}$
- 2nd Regular: $\mathrm{u_{max}} = 7.0655\ \mathrm{m/s}$
- 5th Regular: $\mathrm{u_{max}} = 6.9606\ \mathrm{m/s}$
- Linear Random (No Stretch): $\mathrm{u_{max}} = 3.2598 * 10^{295}\mathrm{m/s}$ (>20m/s, Non-realistic)
- Linear Random (Wheeler): $\mathrm{u_{max}} = 7.2724\mathrm{m/s}$
- Second-Order Random: $\mathrm{u_{max}} = 10.2087\mathrm{m/s}$

4. Discussion

4.1 Regular vs. Random Wave Representation

Forcing an irregular storm sea-state into a regular wave approximation is fundamentally limited. While Stokes' Fifth-Order theory successfully models the asymmetry of a steep wave, it imposes a physically unrealistic periodicity and symmetry onto the profile. The regular wave assumption ignores the randomness of the wave group, which is the primary driver of the largest crest recorded.

4.2 The wave Nature: Under-prediction and Physics

As shown in Table 1, the LRWT model significantly under-predicts the crest height. The Second-Order model reduces this error by introducing sum-frequency interactions that sharpen the crest. However, the Second-Order model likely still exhibits a discrepancy (over-prediction) compared to the recorded wave. The physical limitation is from double counting of second order terms. Applying the second-order model to the full buoy spectrum tends to over-predict crest heights because bound harmonics present in the measured spectrum are incorrectly interpreted as free waves. This leads to double-counting of nonlinear energy and typically leads to over-prediction.

4.3 Kinematics and Design Safety

Utilizing raw Linear Random Wave Theory without stretching results in exponential contamination, yielding velocities that would lead to massive, uneconomical over-engineering.

Wheeler Stretching effectively curtails this blow-up and produces a plausible profile, yet it remains a purely empirical "fudge factor" that strictly violates the Laplace equation and mass continuity. The results are still acceptable although there are errors compared to reality.

The Second-Order Random Wave model provides the most rigorous solution. By explicitly modelling the bound-wave interactions, it naturally restricts high-frequency particle orbits near the crest without empirical stretching. As observed in the results, the 2nd-order velocity profile is generally fuller near the surface than the Wheeler profile, suggesting that Wheeler stretching may slightly under-estimate the kinematic load on the structure's splash zone.

5. Conclusion

Determining wave kinematics is highly sensitive to the chosen framework. Regular wave models fail to capture the chaotic nature of storm seas. While Linear Random Wave Theory successfully reconstructs the general phase of the wave group, it dangerously over-predicts kinematics if unstretched and relies on empirical approximations if stretched.
For the reliable and economic design of offshore infrastructure, **Second-Order Random Wave Theory** is the recommended standard although it has mismatching surface profile, but for safety consideration SORWT is still suggested to design purpose. It captures the essential non-linear sharpening of the crest and provides a physics-based calculation of velocity. However, for large waves of this magnitude, even second-order theory has limits, and designers must account for the potential of third-order modulational instabilities creating crests higher than theoretical predictions.

6. Appendix

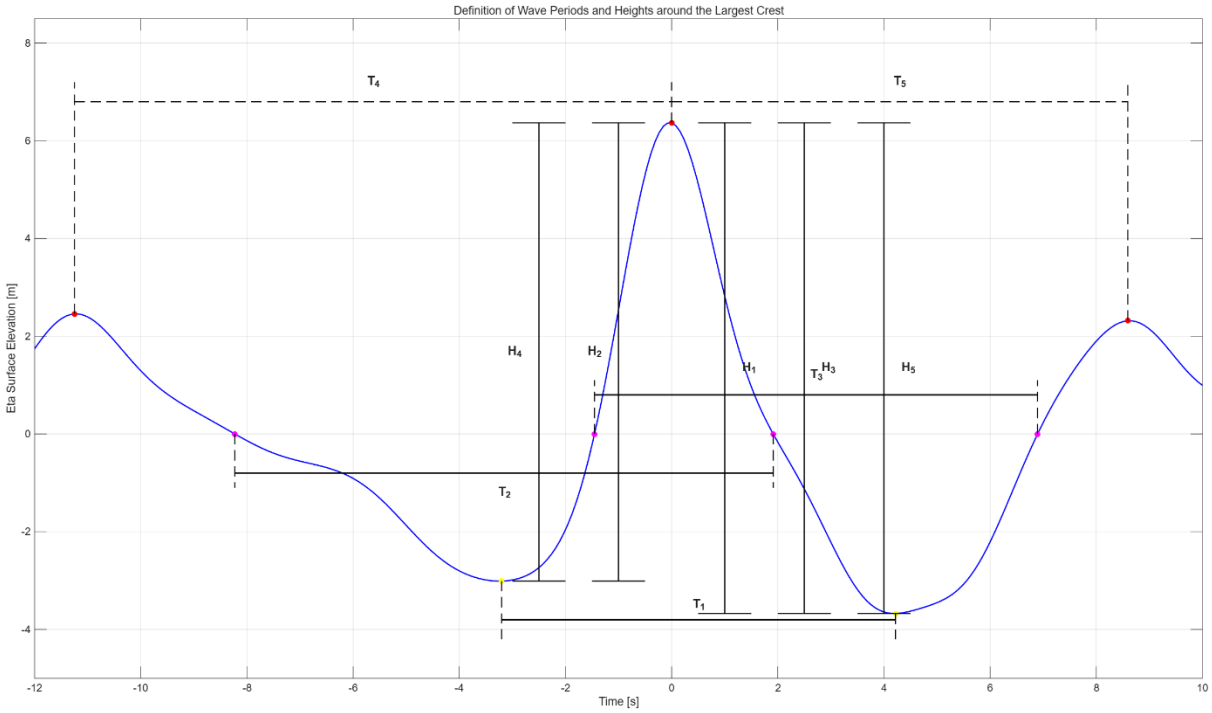


Figure 6.1 T and H indications

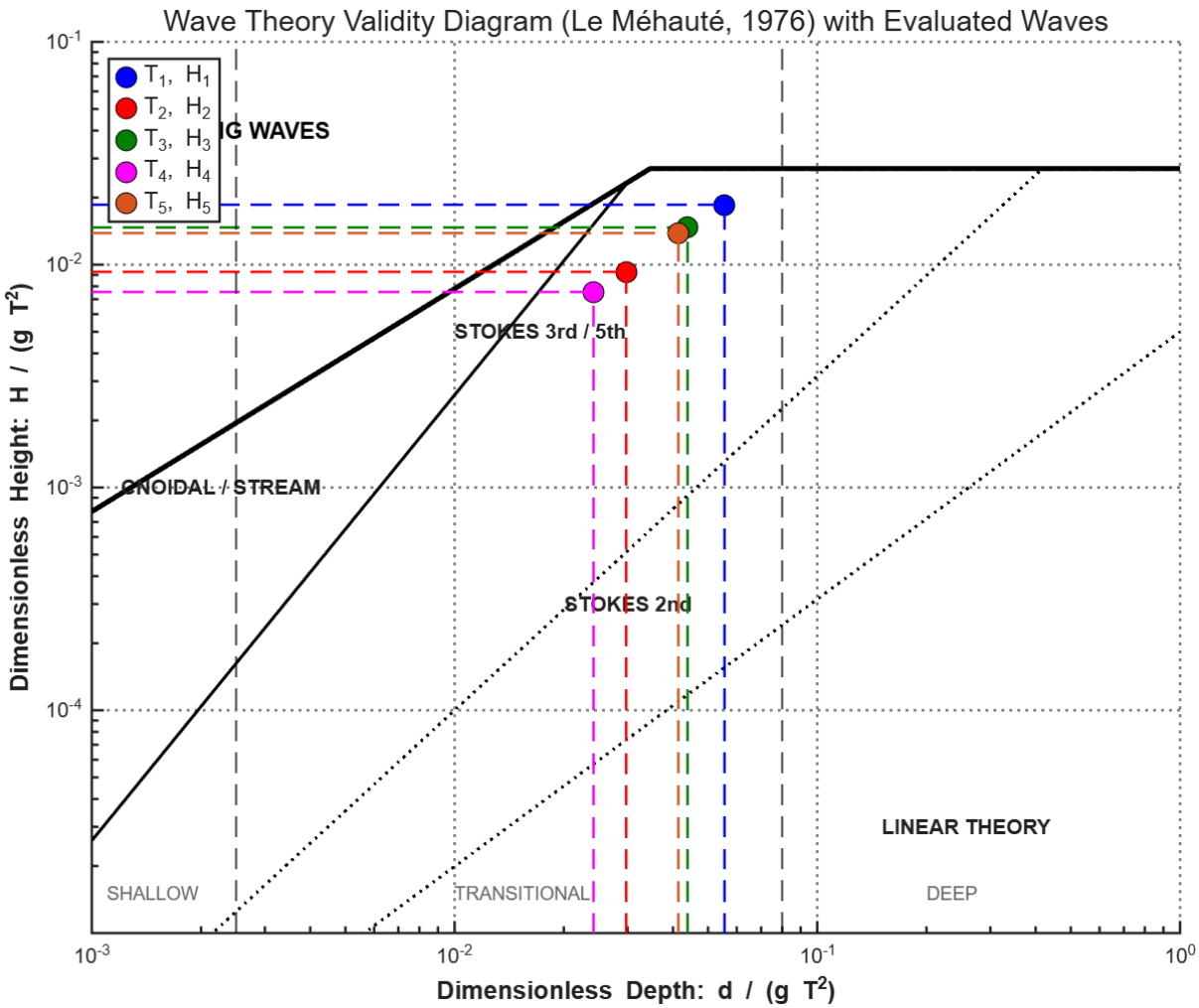


Figure 6.2 Validity diagram, showing the crest fits in Stokes's 5th

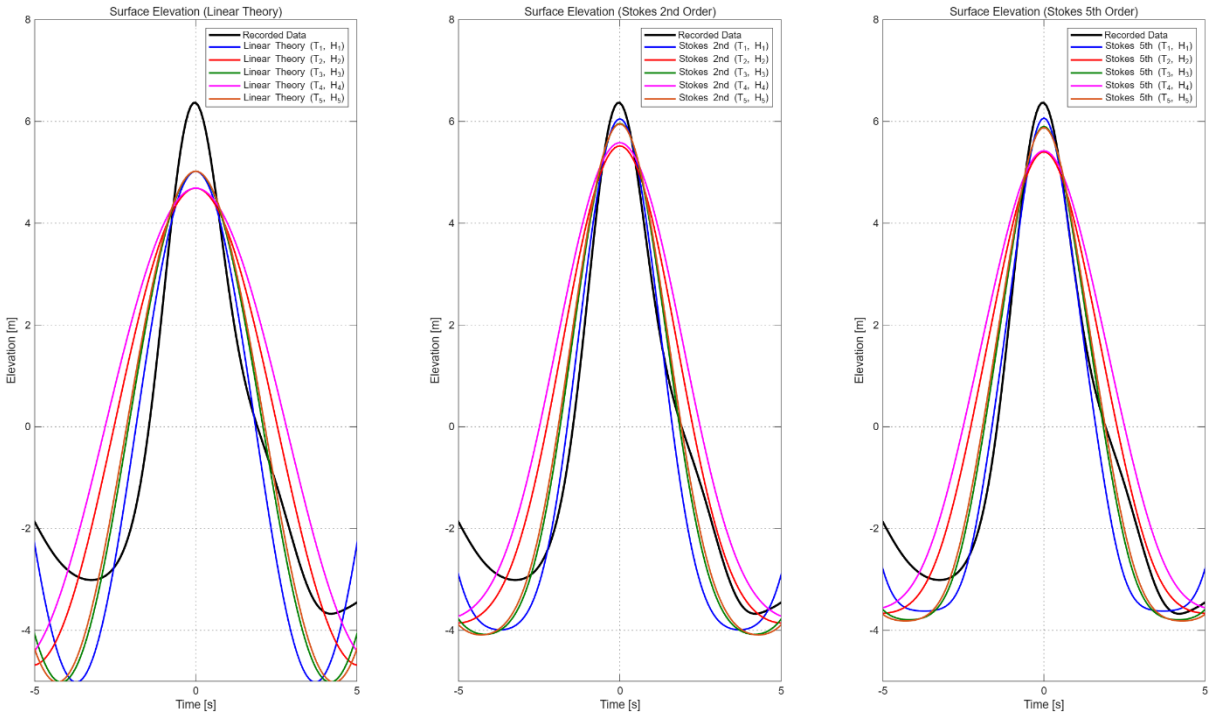


Figure 6.2 crest matching for different T and H

Wave Theory Equations

1. Linear (Airy) Wave Theory

Surface elevation:

$$\eta = \frac{H}{2}\cos(kx - \omega t)$$

Horizontal velocity:

$$u = \frac{H}{2}\omega\frac{\cosh k(z+d)}{\sinh(kd)}\cos(kx - \omega t)$$

Dispersion relation:

$$\omega^2 = gk\tanh(kd)$$

where H is the wave height, k is the wavenumber, ω is the angular frequency, d is the water depth, and z is measured upward from the still water level (SWL).

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2. Second-Order Regular (Stokes 2nd) Wave Theory

Surface elevation:

$$\eta = \frac{H}{2}\cos(kx - \omega t) + \frac{H^2 k}{16}\frac{\cosh(kd)}{\sinh^3(kd)}\big(2 + \cosh(2kd)\big)\cos 2(kx - \omega t)$$

Horizontal velocity:

$$u = \frac{H}{2}\omega\frac{\cosh k(z+d)}{\sinh(kd)}\cos(kx - \omega t) + \frac{3}{16}H^2\omega k\frac{\cosh 2k(z+d)}{\sinh^4(kd)}\cos 2(kx - \omega t)$$

Dispersion relation:

$$\omega^2 = gk\tanh(kd)$$

The dispersion relation remains identical to the linear case; nonlinear corrections to k first appear at fourth order.

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3. Stokes Fifth-Order Wave Theory (Fenton, 1985)

Surface elevation (bed origin, y measured from seabed):

$$k\eta(x) = kd + \varepsilon\cos(kx) + \varepsilon^2 B_{22}\cos(2kx) + \varepsilon^3 B_{31}[\cos(kx) - \cos(3kx)]$$
$$+\varepsilon^4[B_{42}\cos(2kx) + B_{44}\cos(4kx)]$$
$$+\varepsilon^5[-(B_{53} + B_{55})\cos(kx) + B_{53}\cos(3kx) + B_{55}\cos(5kx)]$$

Velocity potential:

$$\phi(x,y) = -cx + c_0\left(\frac{g}{k^3}\right)^{1/2}\sum_{i=1}^{5}\varepsilon^i\sum_{j=1}^{i}A_{ij}\cosh(jky)\sin(jkx)$$

Horizontal velocity (lab frame):

$$u = c_0\sqrt{\frac{g}{k}}\sum_{j=1}^{5} j\ C_j \cosh(jky)\cos(jkx)$$

where the combined coefficients C_j are:

$$C_1 = \varepsilon A_{11} + \varepsilon^3 A_{31} + \varepsilon^5 A_{51}$$

$$C_2 = \varepsilon^2 A_{22} + \varepsilon^4 A_{42}$$

$$C_3 = \varepsilon^3 A_{33} + \varepsilon^5 A_{53}$$

$$C_4 = \varepsilon^4 A_{44}$$

$$C_5 = \varepsilon^5 A_{55}$$

Nonlinear dispersion relation:

$$\omega^2 = gk\tanh(kd)\left[1 + \varepsilon^2 C_1^{(d)} + \varepsilon^4 C_2^{(d)}\right]$$

where $\varepsilon = kH/2$ is the wave steepness parameter, y is measured from the seabed, $S = \text{sech}(2kd)$, and the coefficients A_{ij}, B_{ij} are functions of kd as given in Fenton (1985).

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4. Linear Random Wave Theory

Surface elevation:

$$\eta = \sum_{m=1}^{M} a_m \cos(k_m x - \omega_m t + \alpha_m)$$

Horizontal velocity:

$$u = \sum_{m=1}^{M} a_m\, \omega_m \frac{\cosh k_m(z+d)}{\sinh(k_m d)} \cos(k_m x - \omega_m t + \alpha_m)$$

Dispersion relation (applied independently to each component):

$$\omega_m^2 = g k_m \tanh(k_m d)$$

where a_m, ω_m, k_m, and α_m are the amplitude, angular frequency, wavenumber, and phase of the m-th spectral component, and M is the total number of components.

Wheeler Stretching

The horizontal velocity with Wheeler stretching replaces z with a stretched coordinate z^*:

$$u = \sum_{m=1}^{M} a_m\, \omega_m \frac{\cosh k_m(z^*+d)}{\sinh(k_m d)} \cos(k_m x - \omega_m t + \alpha_m)$$

$$z^* = \frac{z - \eta(t)}{1 + \frac{\eta(t)}{d}}$$

where $\eta(t)$ is the measured wave profile. This maps the instantaneous free surface to $z^* = 0$.

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5. Second-Order Random Wave Theory

Surface elevation:

$$\eta = \eta^{(1)} + \eta^{(2)}$$

First-order component:

$$\eta^{(1)} = \sum_{i=1}^{N_c} a_i \cos(\psi_i)$$

Second-order component:

$$\eta^{(2)} = \frac{1}{4}\sum_{i=1}^{N_c}\sum_{j=1}^{N_c} a_i\, a_j \left\{\left[\frac{D_{ij}^{-} - \left(\mathbf{k}_i \cdot \mathbf{k}_j - R_i R_j\right)}{\sqrt{R_i R_j}} + \left(R_i + R_j\right)\right]\cos\left(\psi_i - \psi_j\right) + \left[\frac{D_{ij}^{+} - \left(\mathbf{k}_i \cdot \mathbf{k}_j + R_i R_j\right)}{\sqrt{R_i R_j}} + \left(R_i + R_j\right)\right]\cos\left(\psi_i + \psi_j\right)\right\}$$

where $\psi_i = k_i x\cos\theta_i + k_i y\sin\theta_i - \omega_i t + \phi_i$ and $R_i = k_i \tanh(k_i d)$.

Horizontal velocity:

$$u = u^{(1)} + u^{(2)}$$

First-order velocity:

$$u^{(1)} = \sum_{i=1}^{N_c} \frac{a_i g k_i \cos\theta_i}{\omega_i \cosh(k_i d)} \cosh[k_i(z+d)]\cos(\psi_i)$$

Second-order velocity (self-interaction):

$$u_{self}^{(2)} = \sum_{i=1}^{N_c} A_1^{(i)} \cdot 2k_i \cos\theta_i \cdot \cosh[2k_i(z+d)]\cos(2\psi_i)$$

Second-order velocity (cross-interaction):

$$u_{cross}^{(2)} = \sum_{i=1}^{N_c}\sum_{j>i}^{N_c} A_2 \left(k_i\cos\theta_i - k_j\cos\theta_j\right)\cosh\left[k_{ij}^{-}(z+d)\right]\cos\left(\psi_i - \psi_j\right)$$

$$+\sum_{i=1}^{N_c}\sum_{j>i}^{N_c} A_3\left(k_i\cos\theta_i + k_j\cos\theta_j\right)\cosh\left[k_{ij}^{+}(z+d)\right]\cos\left(\psi_i+\psi_j\right)$$

Interaction wavenumbers:

$$k_{ij}^{+} = \left|\mathbf{k}_i + \mathbf{k}_j\right| = \sqrt{k_i^2 + k_j^2 + 2k_ik_j\cos\left(\theta_i - \theta_j\right)}$$

$$k_{ij}^{-} = \left|\mathbf{k}_i - \mathbf{k}_j\right| = \sqrt{k_i^2 + k_j^2 - 2k_ik_j\cos\left(\theta_i - \theta_j\right)}$$

Interaction coefficients:

$$D_{ij}^{+} = \frac{\left(\sqrt{R_i}+\sqrt{R_j}\right)\left[\sqrt{R_j}(k_i^2 - R_i^2) + \sqrt{R_i}\left(k_j^2 - R_j^2\right)\right]}{\left(\sqrt{R_i}+\sqrt{R_j}\right)^2 - k_{ij}^{+}\tanh\left(k_{ij}^{+}d\right)} + \frac{2\left(\sqrt{R_i}+\sqrt{R_j}\right)^2\left(\mathbf{k}_i\cdot\mathbf{k}_j - R_iR_j\right)}{\left(\sqrt{R_i}+\sqrt{R_j}\right)^2 - k_{ij}^{+}\tanh\left(k_{ij}^{+}d\right)}$$

$$D_{ij}^{-} = \frac{\left(\sqrt{R_i}-\sqrt{R_j}\right)\left[\sqrt{R_j}(k_i^2 - R_i^2) - \sqrt{R_i}\left(k_j^2 - R_j^2\right)\right]}{\left(\sqrt{R_i}-\sqrt{R_j}\right)^2 - k_{ij}^{-}\tanh\left(k_{ij}^{-}d\right)} + \frac{2\left(\sqrt{R_i}-\sqrt{R_j}\right)^2\left(\mathbf{k}_i\cdot\mathbf{k}_j + R_iR_j\right)}{\left(\sqrt{R_i}-\sqrt{R_j}\right)^2 - k_{ij}^{-}\tanh\left(k_{ij}^{-}d\right)}$$

with $R_i = k_i\tanh(k_id)$.